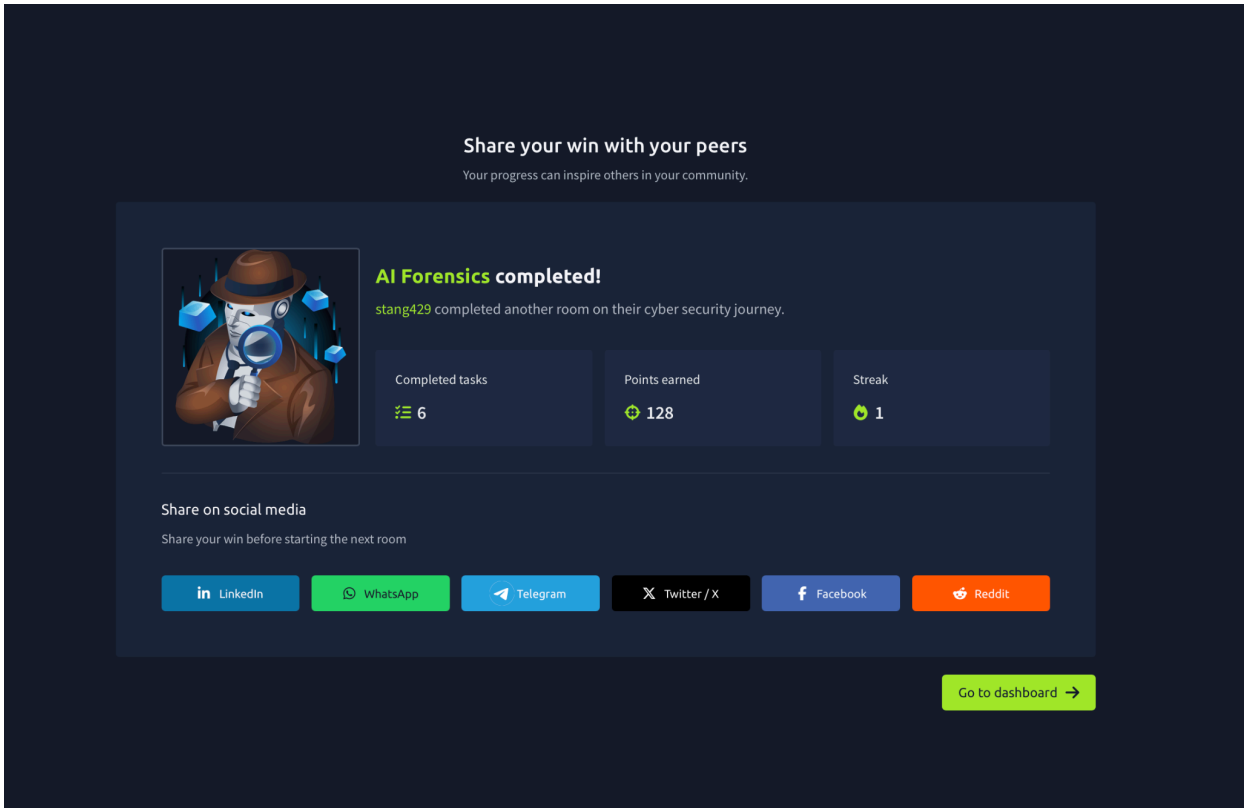


Why can accuracy alone be misleading in a forensic investigation? What is the difference between precision and recall in the context of AI-supported investigation?

Accuracy alone can be misleading in a forensic investigation because it does not account for precision and recall. Accuracy treats all errors as equal, but in forensics they aren't. Missing a key piece of evidence (a false negative) and wrongly flagging an innocent person's file (a false positive) have very different consequences, and which one is worse depends on the stage of the investigation. Low precision means many false alarms, as investigators waste time reviewing irrelevant material, and innocent people may be wrongly implicated. Low recall means missed evidence, where incriminating material may go undetected, and the investigation can reach a wrong or incomplete picture.

Should AI-generated findings be treated as evidence, leads, or conclusions? Explain your answer.

AI-generated findings can be treated as leads, but not without human review, reasoning, and judgment. In general, conclusions require human judgment about context, intent, and the full body of facts. AI models' thinking is often a "black box"; they can't capture the nuances of intent, can't weigh competing explanations, and can be confidently wrong. Their error rates may be unknown or unvalidated for the specific data at hand, and they can carry bias from training data or drift over time. A prime example is how facial recognition technology has many biases inherently. Unexplainable and irreproducible outputs are hard to defend under scrutiny, and the very basis of evidence is reliability and explainability. However, AI is good at surfacing patterns, anomalies, and candidates for review at a large and efficient scale, while potentially flagging possibilities that humans may have not thought of.